Incident Investigation Report

Registry Persistence Activity on DESKTOP-5OG7C28

Date of Activity: 2026-10-27 (UTC)
Host: DESKTOP-5OG7C28 (Windows_VM, 192.168.232.130)
User Context: Darragh Casey (High Integrity, Session 1)
Data Sources: Wazuh Sysmon telemetry, Security Auditing, Kernel-Power events, and Wazuh alerts

1. Executive Summary

On 2026-10-27 around 12:05:07 UTC, a process running as user Darragh Casey (high integrity, interactive session) on host DESKTOP-5OG7C28 executed a chain of commands originating from powershell.exe (PID 12276). This culminated in the creation of a registry Run key entry named “Atomic Red Team” pointing to C:\Path\AtomicRedTeam.exe. The modification was performed via cmd.exe invoking reg.exe and was confirmed by a Sysmon Registry value set (Event ID 13).

Preceding this, the same PowerShell process spawned multiple instances of hostname.exe and whoami.exe from the directory C:\Users\Darragh Casey\Desktop\S_3\. Wazuh alerts fired on the command-prompt execution, the registry modification, and a related process-create rule.

Observed fact: A Run key was written that would cause C:\Path\AtomicRedTeam.exe to execute on the next logon of the affected user.
Inference: This constitutes a persistence mechanism. Confidence in the technical sequence is high; confidence that the activity is unauthorized/malicious is medium (laboratory context and the explicit “Atomic Red Team” naming are noted but the activity is assessed on its own merits).

2. Scope of Activity

All activity is confined to a single Windows host (DESKTOP-5OG7C28 / Windows_VM, IP 192.168.232.130) within an isolated laboratory environment. The relevant window is approximately 12:02–12:05 UTC on 2026-10-27.

Observed activity includes routine system service logons and credential-manager enumerations (early in the window), session unlock and input-related power events, and a single PowerShell process (PID 12276) spawning discovery binaries and then a registry-modification command chain. No network connections, file writes outside the registry action, or additional persistence mechanisms appear in the supplied telemetry. The PowerShell parent process itself has no creation event in the provided logs (it was already running).

3. Detailed Investigation

Directly observed facts

Process tree (all children of PowerShell PID 12276 / ProcessGuid {4b2ce92d-8a8c-6ae0-6206-000000001a00}):

• Multiple hostname.exe and whoami.exe launches (CWD: C:\Users\Darragh Casey\Desktop\S_3\).
• cmd.exe (PID 7912) with command line: "cmd.exe" /c REG ADD "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /V "Atomic Red Team" /t REG_SZ /F /D "C:\Path\AtomicRedTeam.exe" (CWD: Temp directory).
• Child reg.exe (PID 2824) executing the identical REG ADD command.

Sysmon Event ID 13 (Registry value set) at 12:05:07.993Z: TargetObject = HKU\S-1-5-21-2586188580-1629295405-1270050116-1001\Software\Microsoft\Windows\CurrentVersion\Run\Atomic Red Team; Details = C:\Path\AtomicRedTeam.exe; Image = reg.exe; User = DESKTOP-5OG7C28\Darragh Casey; RuleName =

T1060,RunKey.

User context: DESKTOP-5OG7C28\Darragh Casey, LogonId 0xFCD15, TerminalSessionId 1, IntegrityLevel High. Wazuh rules 92052, 92041 and 92302 fired on the cmd/reg/Event-13 sequence.

Inferences

The Run key write is a classic persistence technique that will launch the named executable at next user logon. The preceding hostname/whoami executions are consistent with basic system/user discovery. The value name “Atomic Red Team” and placeholder path C:\Path\AtomicRedTeam.exe are consistent with red-team testing frameworks, but the telemetry itself shows a completed registry persistence action. No evidence of subsequent execution of the payload, privilege escalation beyond the existing high-integrity session, or lateral movement is present.

Irrelevant/benign events (system service logons, MoUsoCoreWorker, CameraMonitor svchost, credential-manager enumerations, Kernel-Power session transitions) were reviewed and excluded from the primary narrative as they do not connect to the PowerShell process tree or the registry modification.

4. Evidence Analysis

The following key records support the findings. All times are UTC on 2026-10-27.

Time (UTC)	Event ID	Key Observation
12:05:06.595–07.504	1 (Sysmon)	Multiple hostname.exe / whoami.exe; Parent PID 12276 powershell.exe; CWD Desktop\S_3
12:05:07.904	1	cmd.exe PID 7912; CommandLine contains REG ADD … “Atomic Red Team” … C:\Path\AtomicRedTeam.exe; Parent powershell.exe
12:05:07.965	1	reg.exe PID 2824; identical REG ADD command; Parent cmd.exe
12:05:07.993	13	SetValue on HKU\…\Run\Atomic Red Team = C:\Path\AtomicRedTeam.exe; Image reg.exe

Hashes, integrity level (High), and session identifiers are consistent across the user-context process creates. The Wazuh alert timestamps align with these Sysmon events. The “Base64-like pattern” alert (rule 92041) fired on the reg.exe create; the actual data written is a plain path, so the rule description does not perfectly match the payload but the underlying action is clear.

5. Timeline of Events

• **12:02:05Z** – System service logons (4624/4672) and credential-manager enumerations (5379) by SYSTEM and user Darragh Casey.
• **12:02:53Z** – MoUsoCoreWorker.exe started by UsoSvc (benign).
• **12:03:53Z** – Kernel-Power session transition (SessionUnlock).
• **12:04:50Z** – Further session/input activity and CameraMonitor svchost start.
• **12:05:06.595Z–07.504Z** – PowerShell PID 12276 spawns hostname.exe (×4) and whoami.exe (×2) from Desktop\S_3.
• **12:05:07.886Z** – PowerShell spawns cmd.exe with REG ADD Run-key command (CWD Temp).
• **12:05:07.953Z** – cmd.exe spawns reg.exe executing the REG ADD.
• **12:05:07.993Z** – Sysmon records SetValue of the Run key entry.
• **12:05:15Z** – Wazuh alerts (92052, 92041, 92302) generated.

6. MITRE ATT&CK; Mapping

Tactic	Technique	Evidence

Execution	T1059.003 Windows Command Shell	cmd.exe spawned by powershell.exe with the REG ADD command line (Sysmon EID 1, alert 92052)
Persistence / Privilege Escalation	T1547.001 Registry Run Keys / Startup Folder	Event ID 13 SetValue on HKCU/HKU \…\CurrentVersion\Run\Atomic Red Team = C:\Path\AtomicRedTeam.exe; alert 92302
Defense Evasion	T1112 Modify Registry	Same registry modification
Discovery	T1033 System Owner/User Discovery; T1082 System Information Discovery	Multiple whoami.exe and hostname.exe children of the same PowerShell process

T1027 (Obfuscated Files or Information) appears in alert 92041 but is not strongly supported by the observed payload (plain path string).

7. Overall Investigation Conclusion

Observed facts (high confidence):

A high-integrity PowerShell process belonging to user Darragh Casey executed discovery binaries and then successfully wrote a registry Run key that will launch C:\Path\AtomicRedTeam.exe on the next logon of that user. The full process tree and the registry write are directly evidenced by Sysmon Events 1 and 13.

Inferences (medium confidence):

The activity implements a persistence mechanism. Given the laboratory setting, the explicit “Atomic Red Team” naming, and the placeholder path, the action is consistent with a controlled detection test. However, the telemetry alone demonstrates a completed Run-key persistence write with no countervailing evidence that the key was subsequently removed or that the activity was authorized within the supplied logs. No additional malicious follow-on activity is visible.

Overall confidence: High in the technical sequence and the existence of the persistence artifact; medium in the determination of malicious intent versus authorized testing.

Recommendation: Verify whether the Run key remains present and whether C:\Path\AtomicRedTeam.exe exists/executes on next logon; review the full PowerShell console history or script that was running under PID 12276 if further context is required.

— End of Report —
Generated from supplied Wazuh/Sysmon telemetry only. Isolated laboratory context acknowledged.